MIL-G-23723(AS)
AMENDMENT-2
1 Aug 1967
Superseding
AMENDMENT-1
15 May 1967

MILITARY SPECIFICATION

Gyroscope, Rate Switching CN-495/AJB-3

This amendment forms a part of Military Specification MIL-G-23723(WEP) dated 15 January 1964 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this Amendment, basic MIL-G-23723(WEP) is changed to MIL-G-23723(AS) and throughout the specification Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

*Page 2, Paragraph 2.1, under SPECIFICATIONS Military: Delete

"MIL-E-4682 Electron Tubes and Transistors, Choice and Application of"

"MIL-R-23094 Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for"

Page 3, Paragraph 2.1, under STANDARDS, Federal: Add

FED-STD-209 Clean Room and Work Station Requirements, Controlled Environment

Page 3, Paragraph 2.1, under STANDARDS Military: Add

MIL-STD-781 Reliability Test, Exponential Distribution

Page 5, Paragraph 3.4: Delete and Substitute

"3.4 Design and Construction - The Rate Switching Gyroscope shall provide switching during turns within specified time intervals at specified turn rates. Design shall provide no switching within specified time intervals after return to zero turn rate from a particular rate of turn. The Rate Switching Gyro shall consist of three assemblies:

1. The Gyroscope Assembly
2. The Hydraulic Viscous Damping Assembly
3. All Electronics Assembly

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Each assembly shall be separate, independent and concentric with the other. The Hydraulic Damping Medium shall conform with the requirements of 3.4.4.2. The Gyroscope Assembly, if practical, shall conform with filling medium requirements of 3.4.4.1. The design shall have a Hydraulic Viscous Damping in the dash pot mechanism which shall always be free from hangup. The Rate Switching Gyro shall operate normally when properly connected in the gyro platform system conforming to the requirements and specifications in either right side up or inverted positions."

Page 6, Paragraph 3.4.4.1: Delete and Substitute

"3.4.4.1 Filling Medium - The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point - 65°C) at the filling pressure. The filling medium shall be a mixture of 95 percent helium and 5 percent carbon dioxide, or a filling medium approved by the procuring activity. The absolute pressure of the filling medium of the filling case shall be approximately one atmosphere.

Page 6: Add

"3.4.4.2 Hydraulic Medium - The Hydraulic Filling Medium shall be Dow Corning 510 fluid or equivalent and shall be filtered so that the maximum particle size shall not exceed 8.8 Microns diameter in the medium."

*Page 6, Paragraph 3.5.1(1): Delete

"Three Phase" and substitute "Single Phase"

*Page 6, paragraph 3.5.3: Delete

Page 6: Add

"3.5.6 Switching Mechanism - The switching mechanism shall contain a single pole double throw switch with solid platimum alloy contact points or equivalent. The normally open circuit shall close and the normally closed circuit shall open when a turn rate of 15° per minute or more is imposed on the Gyroscope. The contact rating shall be not less than 2 Amperes resistive load at 115 volts. The contact circuits shall be isolated from the other circuitry in the gyroscope and from the case."

Page 11, Paragraph 4.4.2.2.1: Delete

"Test for Helium"

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Page 11, Paragraphs 4.4.3 through 4.4.3.5: Delete and Substitute

"4.4.3 Reliability Procedure - Reliability Assurance-Testing will be accomplished through a production burn-in. Each equipment produced under this contract shall be operated for one hundred-fifty (150) operating hours burn-in, and the test data (operating hours and failures) shall be compared to the accept/reject criteria of Test Plan V of MIL-STD-781. Delivery of equipment shall not be held up to await a decision under the Test Plan. In event an accept decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. In the event the accumulation of time from the 100 hour burn-in, results in the second accept decision, the test duration on each equipment shall be reduced to 50 hours. At any time that the current totals of test hours and test failures plotted on Test Plan V curves show a reject situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the Contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

"4.4.3.1 Test Level - The Test Level shall be "F" from MIL-STD-781.

4.4.3.2 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.6.

4.4.3.3 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

1. Whenever performance characteristics fall below the acceptance requirements (4.4.3.2), at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure.

2. If more than one item is responsible for degrading the system, each item shall be counted a failure."

Page 16, Paragraph 4.7: Delete "Signal Adapters" and Substitute "Rate Switching Gyros"

Page 16, Paragraph 4.6.20: Delete

Page 17, Paragraph 6.1 Delete and Substitute

"6.1 Intended Use - The Rate Switching Gyroscope covered by this specification is intended for use in aircraft as a unit of the AN/AJB-3, AN/AJB-3A and AN/ASN-50.

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Page 17: Add

"6.6 Performance Characteristics to be Measured - At least twice each week, the Rate Switching Gyroscope shall be subjected to and meet the requirements of the following tests:

1. Switching (4.6.3)
2. Yaw Oscillation (4.6.4)
3. Steady State Turn and Yaw Oscillation (4.6.5)*

An asterisk preceding a change to the specification denotes a change which did not appear in Amendment-1

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